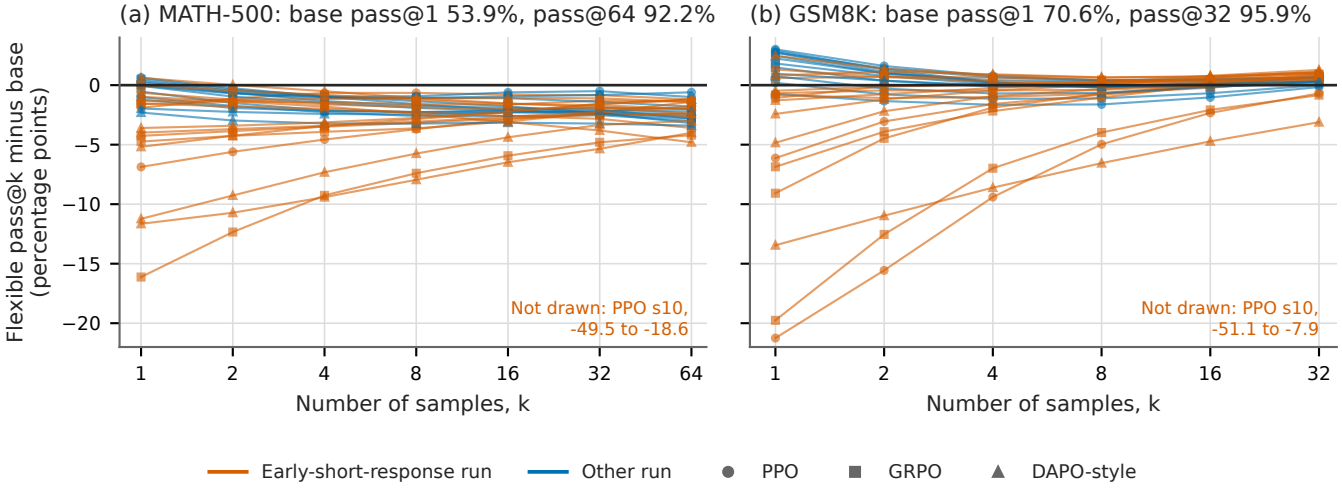
(a) MATH-500: base pass@1 53.9%, pass@64 92.2%
(b) GSM8K: base pass@1 70.6%, pass@32 95.9%
Flexible pass@k minus base
(percentage points)
0
−5
−10
−15
−20
Not drawn: PPO s10,
-49.5 to -18.6
Not drawn: PPO s10,
-51.1 to -7.9
1
2
4
8
16
32
64
Number of samples, k
Early-short-response run
Other run
PPO
GRPO
DAPO-style